

+

×

—

÷

DENSITY OPERATOR

$$\rho = \sum_n p_n |\psi_n\rangle\langle\psi_n|$$

$$\langle A \rangle = \text{Tr}(\rho A)$$

$$\rho = \sum_{n,m} \rho_{nm} |n\rangle\langle m|$$

$$\rho_{nm} = \langle n | \rho | m \rangle$$

PROBABILITY THAT A BASIC MEASUREMENT IN THIS BASIS WOULD YIELD THE RESULT n

IN A SUPERPOSITION OF BASIC STATES, THERE IS A DEFINITE PROBABILITY RELATION

$$|m_1\rangle = \frac{1}{\sqrt{2}}(|z_+, z_+\rangle + |z_-, z_-\rangle)$$

$$\bullet |m_2\rangle = e^{i\pi/3} \frac{1}{\sqrt{2}}(|z_+, z_-\rangle + |z_-, z_+\rangle)$$

$$\bullet |m_3\rangle = \frac{1}{\sqrt{2}}\left(e^{i\pi/4}|z_+, z_+\rangle + e^{i\pi/4}|z_-, z_-\rangle\right)$$

$$A = \sum_{\alpha} \alpha |\alpha\rangle\langle\alpha| \quad \rho_i = |m_i\rangle\langle m_i| \quad i=1,2 \text{ OR } 3$$

DENSITY OPERATOR

$$\text{Tr}(A\rho_1) = \text{Tr}\left(\sum_{\alpha} \alpha |\alpha\rangle\langle\alpha| |m_1\rangle\langle m_1|\right)$$

$$= \text{Tr}\left(\sum_{\alpha} \alpha \langle\alpha|m_1\rangle | \alpha\rangle\langle m_1| \right)$$

$$= \sum_n \langle n | \left(\sum_{\alpha} \alpha \langle\alpha|m_1\rangle | \alpha\rangle\langle m_1| \right) | n \rangle$$

$$= \sum_{\alpha} \alpha \langle\alpha|m_1\rangle \sum_n \langle n | \alpha\rangle\langle m_1| n \rangle$$

ARE NUMBERS

$$= \sum_{\alpha} \alpha \langle\alpha|m_1\rangle \langle m_1 | \left(\sum_n |n\rangle\langle n| \right) | \alpha \rangle$$

1

$$= \sum_{\alpha} \alpha |\langle\alpha|m_1\rangle|^2$$

SAME EXPECTATION VALUE

$$|\langle\alpha|m_1\rangle|^2 \leftarrow$$

$$|\langle\alpha|m_2\rangle|^2 = |\langle\alpha|e^{i\pi/3}m_1\rangle|^2 = |e^{i\pi/3}|^2 |\langle\alpha|m_1\rangle|^2$$

$$|\langle\alpha|m_3\rangle|^2 \quad \downarrow \quad 1$$

Any physical process which has the effect of suppressing the coherence of the density matrix (in some particular basis) is called a **decoherence process**

Exercise 8.12 Consider the qubit state with density matrix

$$\rho_K = \frac{1}{2} \begin{pmatrix} 1 & K \\ K & 1 \end{pmatrix}$$

where $0 \leq K \leq 1$. A measurement of X is made on this qubit. Calculate $|P(+1) - P(-1)|$ as a function of K

$$\mathcal{B} = \{|0\rangle, |1\rangle\}$$

$$\rho_K = \frac{1}{2} \begin{pmatrix} 1 & K \\ K & 1 \end{pmatrix}$$

$$|P(+1) - P(-1)|$$

← expectation of X

X is Pauli matrix

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

1) $\langle X \rangle = \text{Tr}(\rho_K X)$

$$= \text{Tr} \left(\frac{1}{2} \begin{pmatrix} 1 & K \\ K & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) = \frac{1}{2} \text{Tr} \begin{pmatrix} K & 1 \\ 1 & K \end{pmatrix} = \frac{1}{2} (2K) = K$$

SUPPOSE $K=0$

$$\rho_0 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \langle X \rangle = 0$$

2) $P(+1)$

$$\langle X+ | \rho_K | X+ \rangle$$

$$|X_+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \frac{1}{2} \begin{pmatrix} 1 & K \\ K & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1+K \\ K+1 \end{pmatrix} = \frac{1}{4} (1+K + K+1) = \frac{1}{2} (K+1)$$

BRN OPERATOR $K \in \mathbb{R}$

IT IS A PROB, SINCE BY BETWEEN [0,1]

$$P(+1) = \frac{1}{2} (K+1)$$

→ IF $K=1 \Rightarrow P(+1)=1 \rightarrow$ CERTAIN TO FIND $(+1)$

$P(+1)$ STRATEGY 2 TO FIND IT

$$P(+1) = \text{Tr}(\rho_K \rho_{X+})$$

$$\rho_{X+} = |X_+\rangle \langle X_+|$$

$$\downarrow$$

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$|X_+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$|X_+\rangle \langle X_+| = \frac{1}{2} (|0\rangle + |1\rangle) \frac{1}{\sqrt{2}} (\langle 0| + \langle 1|)$$

$$= \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 1|)$$

$$= \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$P(+1) = \text{Tr} \left(\frac{1}{2} \begin{pmatrix} 1 & k \\ k & 1 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right) = \frac{1}{4} \text{Tr} \begin{pmatrix} 1+k & 1+k \\ 1+k & 1+k \end{pmatrix} = \frac{1}{2} (1+k)$$

$$P(-1) = \text{Tr} (P_k P_{\lambda_-})$$

$$|\lambda_-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \quad \rightarrow \quad \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$P_{\lambda_-} = |\lambda_-\rangle \langle \lambda_-| = \frac{1}{2} [|0\rangle\langle 0| + |1\rangle\langle 1| - |0\rangle\langle 1| - |1\rangle\langle 0|]$$

$$P(-1) = \text{Tr} \left(\frac{1}{2} \begin{pmatrix} 1 & k \\ k & 1 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right) = \frac{1}{4} \text{Tr} \begin{pmatrix} 1-k & 1-k \\ 1-k & 1-k \end{pmatrix} = \frac{1}{4} \cdot (1-k+1-k) = \frac{1}{2} (1-k)$$

NOT CARE ABOUT THIS

$$|P(+1) - P(-1)| = \left| \frac{1}{2} (1+k) - \frac{1}{2} (1-k) \right| = \frac{1}{2} |2k| = |k|$$

↓
VISIBILITY

$|\psi^{(RQ)}\rangle$ state of the subsystem R and Q

CONSIDER THE CONDITIONAL STATE OF Q WHEN THE RESULT OF A MEASUREMENT ON R GIVES THE OUTCOME k

$$\sqrt{p_k} |\psi_k^{(Q)}\rangle = \langle k^{(R)} | \psi^{(RQ)} \rangle$$

$$\begin{aligned} p^{(Q)} &= \sum_k p_k |\psi_k^{(Q)}\rangle \langle \psi_k^{(Q)}| \\ &= \sum_k \langle k^{(R)} | \psi^{(RQ)} \rangle \langle \psi^{(RQ)} | k^{(R)} \rangle \end{aligned}$$

SINCE

$$p^{(Q)} = |\psi^{(Q)}\rangle \langle \psi^{(Q)}|$$

FORMERLY ONE CAN WRITE

$$p^{(Q)} = \sum_k \langle k^{(R)} | p^{(RQ)} | k^{(R)} \rangle$$

NOTE: RESULT IS THE SAME REGARDLESS OF THE OBSERVABLE CONSIDERED ON R. WE ARE INDEED MAKING A "POSITIVE TRACE" ON R

GIVEN A GENERIC OPERATOR

$$G^{(Q,R)} = |\alpha^{(R)}\rangle \langle \alpha^{(R)}| \otimes |\beta^{(Q)}\rangle \langle \beta^{(Q)}| = |\alpha^{(R)}\rangle \langle \alpha^{(R)}| \otimes |\beta^{(Q)}\rangle \langle \beta^{(Q)}|$$

$$\text{Tr}_R G^{(Q,R)} = \sum_n \langle n^{(R)} | G^{(Q,R)} | n^{(R)} \rangle$$

$$= \sum_n \underbrace{\langle n^{(R)} | \alpha^{(R)} \rangle \langle \beta^{(Q)} |}_{\text{}} \underbrace{\langle \beta^{(Q)} | n^{(R)} \rangle}_{\text{}} \otimes |\beta^{(Q)}\rangle \langle \beta^{(Q)}|$$

$$= \sum_n \langle \beta^{(Q)} | n^{(R)} \rangle \langle n^{(R)} | \alpha^{(R)} \rangle \otimes |\beta^{(Q)}\rangle \langle \beta^{(Q)}|$$

= MAGIC

$$\Rightarrow \sum_n \underbrace{|h^{(R)} \times h^{(R)}|}_{\frac{1}{2}}$$

$$\rho^{(A)} = \rho_{A \in 0} | \psi^{(A)}_{A \in 0}$$

$$\rho^{(A)} = \sum_K p_K | \psi_K^{(A)} \rangle \langle \psi_K^{(A)} |$$

$$\rho^{(QR)} = | \psi^{(QR)} \rangle \langle \psi^{(QR)} | \quad \rho^{(A)} = \sum_K \langle \psi_K^{(R)} | \rho^{(QR)} | \psi_K^{(R)} \rangle$$

GIVE A GENERIC OPERATOR

$$G^{(QR)} = | \alpha^{(R)} \rangle \langle \beta^{(R)} | \otimes | \gamma^{(Q)} \rangle \langle \delta^{(Q)} |$$

WE DEFINE THE PARTIAL TRACE ON R

$$\text{THUS WE DEFINE } \rho^{(A)} = \text{Tr}_R(\rho^{(QR)})$$

CASE 1: SEPARABLE PURE STATE

$$\rho^{(QR)} = | \alpha^{(R)} \rangle \langle \alpha^{(R)} | \otimes | \gamma^{(Q)} \rangle \langle \gamma^{(Q)} | = | \alpha^{(R)} \rangle \langle \alpha^{(R)} | \otimes | \gamma^{(Q)} \rangle \langle \gamma^{(Q)} |$$

$$\rho^{(A)} = \text{Tr}_R(\rho^{(QR)}) = \langle \alpha^{(R)} | \alpha^{(R)} \rangle | \gamma^{(Q)} \rangle \langle \gamma^{(Q)} | = | \gamma^{(Q)} \rangle \langle \gamma^{(Q)} |$$

$$\text{Tr}_R(\rho^{(QR)}) = \sum_n p_n | \varphi_n^{(A)} \rangle \langle \varphi_n^{(A)} | \quad \sum_i | \omega_i |^2 = 1$$

AS EXPECTED IN THE CASE OF ENTANGLED SUBSYSTEM, THE RESULT

Exercise 8.4 Consider the entangled states

$$|\phi_+\rangle = \frac{1}{\sqrt{2}}(|0,0\rangle + |1,1\rangle) \quad |\psi_-\rangle = \frac{1}{\sqrt{2}}(|1,0\rangle - |0,1\rangle)$$

for a pair of qubits. In each case, find density operators to describe the subsystem states of each qubit. Do the subsystem density operators completely determine the joint state of the composite system?

$$\mathcal{B} = \{|0,0\rangle, |0,1\rangle, |1,0\rangle, |1,1\rangle\}$$

$$\rho^{QR} = \frac{1}{2} \left[\begin{smallmatrix} QR & QR \\ 1 & 0 & 0 & 0 \end{smallmatrix} + \begin{smallmatrix} QR & QR \\ 1 & 1 & 1 & 1 \end{smallmatrix} + \begin{smallmatrix} QR & QR \\ 0 & 0 & 1 & 1 \end{smallmatrix} + \begin{smallmatrix} QR & QR \\ 1 & 1 & 0 & 0 \end{smallmatrix} \right]$$

$$\text{Tr}_R(\rho^{QR}) = \langle 0^R | \rho^{QR} | 0^R \rangle + \langle 1^R | \rho^{QR} | 1^R \rangle$$

$$= \frac{1}{2} \left[\begin{smallmatrix} Q \\ 1 & 0 & 0 & 0 \end{smallmatrix} + \begin{smallmatrix} Q \\ 1 & 1 & 1 & 1 \end{smallmatrix} \right]$$

$$\mathcal{B} = \{|0^Q\rangle, |1^Q\rangle\}$$

$$\rho^Q = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Tr}_Q(\rho^{QR}) \quad \rho^{(R)} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathcal{B} = \{|0^R\rangle, |1^R\rangle\}$$

$$|\psi_-\rangle = \frac{1}{\sqrt{2}}(|1,0\rangle - |0,1\rangle) \quad \rho^{RQ} = \frac{1}{2} \left[\begin{smallmatrix} RQ \\ 1 & 0 & 0 & 0 \end{smallmatrix} + \begin{smallmatrix} RQ \\ 0 & 1 & 0 & 0 \end{smallmatrix} - \begin{smallmatrix} RQ \\ 1 & 0 & 1 & 0 \end{smallmatrix} - \begin{smallmatrix} RQ \\ 0 & 1 & 0 & 1 \end{smallmatrix} \right]$$

↳ FIRST ONE IS R

Q R

$$\text{Tr}_R(\rho^{RQ}) = \langle 0^R | \rho^{RQ} | 0^R \rangle + \langle 1^R | \rho^{RQ} | 1^R \rangle$$

↳ SECOND ONE IS Q

$$= \frac{1}{2} \left[\begin{smallmatrix} Q \\ 1 & 1 & 1 & 1 \end{smallmatrix} + \begin{smallmatrix} Q \\ 0 & 0 & 0 & 0 \end{smallmatrix} \right]$$

$$\rho^Q = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Tr}_Q(\rho^{RQ}) = \rho^R = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

